Chapter 11

All-ceramic crowns

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INTRODUCTION

All-ceramic crowns developed from a desire to restore heavily broken down anterior teeth to a form and function that was aesthetically pleasing. Prior to the development of contemporary tooth-coloured direct restorative materials, and in place of gold or amalgam restorations, anterior teeth could be restored using all-ceramic restorations.

THE PORCELAIN JACKET CROWN

In initial attempts to make all-ceramic restorations, anterior crowns used porcelain with a relatively high concentration of feldspar (a mixture of sodium and potassium aluminosilicates). Although this so-called feldspathic porcelain produced acceptable aesthetic results, the slow propagation of cracks between flaws within the porcelain during function, and also the phenomenon of stress corrosion that arises as a result of hydrolysis of the Si-O groups of the material under favourable alkaline environmental conditions, meant that such crowns could be used only to restore anterior teeth subject to minimal occlusal loading.

Use of this type of crown elsewhere in the mouth simply resulted in premature crown fracture. As a consequence, a considerable amount of research and development has since been undertaken to improve the reliability of dental porcelain to render it suitable for use in anterior and posterior dental restorations. This has centred upon methods of restoration manufacture and the chemical composition of the porcelain. These frequently linked developments are reviewed throughout this chapter and will cover aspects of sintering, casting, hot pressing and injection moulding, and milling.

The addition of alumina to feldspathic porcelain was reported by McLean and Hughes in 1965 and resulted in much stronger dental porcelain which was more resistant to crack propagation. However, aluminous porcelain does not have the same aesthetic qualities as feldspathic porcelain; it does not have the same translucency and cannot reproduce the life-like illusion of a natural tooth crown. Instead, aluminous porcelain can be used to form a coping over the crown preparation (Figure 11.1), which is itself covered with more aesthetic feldspathic porcelain. Using this principle, the first widely used all-ceramic

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Figure 11.1 Alumina coping to be covered with feldspathic porcelain.

crown was developed and was often referred to as the porcelain jacket crown (PJC). This development increased the strength of the PJC to around 120–150 MPa by reducing the likelihood of crack propagation.

The PJC was widely used to provide an aesthetic restoration for upper anterior teeth. However, it was still not strong enough to resist much occlusal loading without the crown breaking (Figure 11.2). Its use was, therefore, confined mainly to upper incisors, and some premolar teeth in minimal occlusal function. This type of crown is not indicated for molar teeth.

The tooth preparation for and construction of a PJC is described in a stylized diagram in Figure 11.3. The crown preparation requires a shoulder margin all around the gingival aspect of the preparation with an axial reduction of approximately 1.0–1.5 mm. Reduction at the incisal edge is in the order of 1.5–2.0 mm with 1.0–1.5 mm interocclusal clearance required.



Figure 11.2 Palatal fracture of a porcelain jacket crown (PJC) restoration on the upper left central incisor due to excessive occlusal loading. These teeth were crowned to mask severe tetracycline staining.

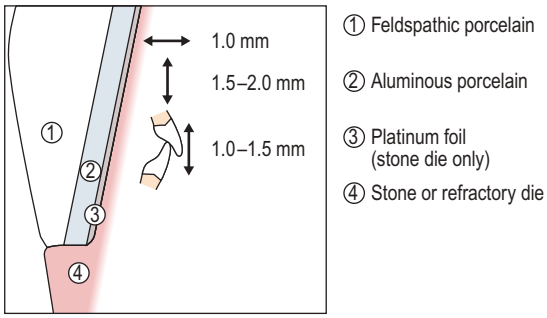


Figure 11.3 Schematic diagram of a porcelain jacket crown: preparation and anatomy.

Fabrication

Traditionally, the impression was cast and a die of the preparation poured using die-stone. A platinum foil matrix was then swaged over the die and an aqueous slurry of aluminous porcelain powder placed over the foil to form the coping. The foil supports the aluminous porcelain slurry in the furnace and may be removed once the crown has been fired and completed. The space left by the platinum foil effectively acts as a die spacer for the luting cement. In order to reduce porosity and shrinkage, firing takes place in a porcelain furnace under vacuum. The process in which the ceramic particles are fused together under heat in this way is called sintering. The aluminous coping is now ready for veneering with feldspathic porcelain which in turn is sintered.

Instead of a stone die model, refractory material may be used which maintains its dimensional stability when subjected to the heat of the porcelain furnace (see Chapter 12 on ceramic veneers). It has been argued that the use of a refractory die results in a more accurate fit of the final restoration to the prepared tooth.

The tooth preparation for the PJC has become a template for modern all-ceramic crowns with one notable exception: the margin has now become a large chamfer or a rounded shoulder. Furthermore, some of the latest all-ceramic crowns now have sufficient strength with an axial tooth reduction of as little as 0.6 mm.

The PJC remained a very popular and widely used restoration for many years. However, there was concern over the bulk of tooth tissue reduction and the inability to withstand occlusal loading. To address these concerns, research concentrated on:

- Decreasing the bulk of tooth tissue reduction required to place an all-ceramic crown
- Development of additional glass ceramic materials and processes by which they can be manipulated
- The use of computer-aided design and computer-aided manufacture (CAD-CAM) and the development of glass ceramic materials with significantly increased strength.

There is now scope to use all-ceramic crowns on any tooth and within increasingly challenging occlusal environments.

DECREASING TOOTH TISSUE REDUCTION FOR AN ALL-CERAMIC CROWN

The dentine-bonded crown

With the advent of adhesively retained porcelain laminate veneers in the 1980s and 1990s, it was a natural progression to extend the preparation to cover the whole surface of the crown, and thus the dentine-bonded crown concept was developed. The strength of this restoration is developed once it is bonded, using a composite resin luting cement, to the underlying tooth structure or composite core. Prior to bonding, the dentine-bonded crown is very fragile and should be treated in the same way as a porcelain laminate veneer.

The use of dentine-bonded crowns is mainly for anterior teeth where occlusal loading is relatively low (Figure 11.4). Such restorations are contraindicated for those patients with an obvious bruxing habit. Application of this type of restoration for premolar and molar teeth, which are normally subject to higher occlusal loading, should be made with caution and after careful examination of the patient's occlusion. Generally, such applications should be avoided. The patient seen in Figure 11.5 has four posterior dentine-bonded crowns chosen for optimum aesthetics as the patient was young. No signs or symptoms of bruxism were noted and canine guidance was achieved; however, despite this, the lower crown fractured and had to be replaced with a metal–ceramic crown.

Tooth preparation for dentine-bonded crowns is kept as minimal as possible and less than that required for metal–ceramic crowns or a traditional PJC. The preparation, in some instances, can be confined to enamel. However, the ceramic should be sufficiently thick to mask discoloured teeth prior to cementation. A minimal shoulder or, more often, a minimal chamfer is the restoration margin of choice. All margins should be supragingival wherever possible to avoid the problems of moisture control at cementation. The axial reduction is in the order of 0.5 mm while the occlusal reduction is between 1.0 and 1.5 mm, with at least 1.0 mm reduction in all excursive movements associated with the preparation (Figure 11.6).

In common with other all-ceramic crown preparations, line and point angles should be rounded to avoid stress concentrations within the porcelain. Figures 11.7–11.9 show the tooth preparation that was carried out for the patients seen in Figures 11.4 and 11.5. Marked palatal erosion (Figure 11.7) has led to exposure of the tertiary (reactionary) dentine that has formed. The only preparation carried out palatally is the cervical chamfer; the tooth wear has removed the rest of the palatal tooth tissue (Figure 11.7). Interocclusal clearance has been created following an



Figure 11.5 Dentine-bonded crowns made from feldspathic porcelain on the upper right premolar and first molar and lower right first molar teeth.



Figure 11.4 Dentine-bonded crowns made from feldspathic porcelain on all six upper anterior teeth.

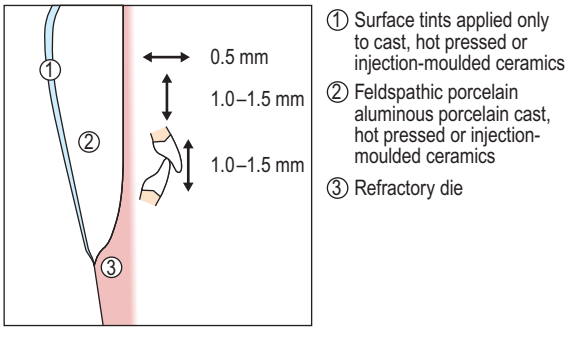


Figure 11.6 Schematic diagram of a dentine-bonded crown: preparation and anatomy.



Figure 11.7 Occlusal view of dentine-bonded crown preparations. The majority of palatal tooth removal has been a result of erosion. The completed result is shown in Figure 11.4.



Figure 11.8 Labial view of anterior dentine-bonded crown preparations. The completed case is shown in Figure 11.4.



Figure 11.9 Buccal view of dentine-bonded crown preparations for posterior teeth with composite resin core augmentations. The completed restorations are shown in Figure 11.5.

increase in the patient's occlusal vertical dimension by placing gold onlays on some of the posterior teeth (Figure 11.8).

The dentine-bonded crown may be made in a variety of ways using low-fusing feldspathic porcelains, alumina-reinforced porcelains or leucite-reinforced porcelains, all on refractory dies. In addition, castable glass ceramics such as Dicor, as well as hot-press and injection-moulding techniques (e.g. IPS Empress 2), may be used to produce such restorations.

As with a ceramic veneer (see Chapter 12), the fitting surface of a feldspathic porcelain dentine-bonded crown is etched in the laboratory using hydrofluoric acid. Once the fit of the crown has been checked in the mouth, the fitting surface should be cleaned and dried, and a silane coupling agent applied to further enhance adhesion of the resin luting cement. In turn, the tooth preparation is etched, rinsed, dried and bond applied to ensure adhesion of the luting cement to the tooth.

Due to the fragility of this type of restoration when made using feldspathic porcelain, it is recommended that occlusal evaluation and any adjustment be completed once the crown has been cemented. Once cemented, the porcelain is bonded to the tooth structure, each component – crown and core – working synergistically together. The result is a restoration where catastrophic crack propagation and failure is less likely to occur. Because the bonding of this type of restorations is so critical any artificial core material used in conjunction with it must be made of composite (Figure 11.9).

DEVELOPMENT OF ADDITIONAL GLASS CERAMIC MATERIALS AND PROCESSES BY WHICH THEY CAN BE MANIPULATED

The development of glass ceramic materials is summarized in Table 11.1, which has been compiled from manufacturers' data, summarized according to the method of manufacture and the clinical applications of these techniques. Some materials are no longer available (see table for details) but are described here for completeness and because history has a tendency to repeat itself in various guises. Their inclusion also illustrates some of the problems experienced and how they have been overcome.

Castable glass ceramic

The development of a castable glass ceramic meant that the lost wax process could be adapted for use with aesthetic restorations (see Chapter 9 for the lost wax technique). Dicor (Dentsply International) was a 70% tetrasilic fluormica glass within a 30% glass matrix which was cast into the space left after vaporization of a wax

Table 11.1 Clinical applications of porcelain according to manufacturing method of restoration

MANUFACTURING METHOD	AVAILABLE MATERIALS	CLINICAL APPLICATION
Sintering	Feldspathic porcelain, aluminous porcelain	Porcelain jacket crowns, veneers. With appropriate alloy core/framework; dentine-bonded crowns and bridges in both anterior and posterior regions of the mouth
	In-Ceram Alumina (Al_2O_3) (VITA, Zahnfabrik, Germany)	Crowns
	In-Ceram Spinell ($MgAl_2O_4$) (VITA, Zahnfabrik, Germany)	Crowns
	In-Ceram Zirconia (VITA, Zahnfabrik, Germany)	Veneers, crowns and anterior bridges
Casting	Dicor (Dentsply International, USA) (*No longer available*)	Anterior crowns
Hot pressing and injection moulding	IPS Empress (Ivoclar Vivadent, Liechtenstein) (Leucite reinforced glass ceramic)	Anterior onlays and crowns
	IPS Empress 2 (Ivoclar Vivadent, Liechtenstein) (Lithium disilicate glass ceramic)	Anterior and posterior onlays and crowns
	IPS e.max Press (Ivoclar Vivadent, Liechtenstein) (Lithium disilicate glass ceramic)	Anterior to premolar onlays and crowns; anterior 3 unit bridgework
Machining	VITABLOCS Mark II (VITA, Zahnfabrik, Germany) (Feldspathic porcelain)	Crowns, veneers
	VITA TriLuxe Bloc (VITA, Zahnfabrik, Germany) (Feldspathic porcelain)	Anterior crowns, veneers
	In-Ceram Alumina (Al_2O_3) (VITA, Zahnfabrik, Germany)	Crowns
	In-Ceram Spinell ($MgAl_2O_4$) (VITA, Zahnfabrik, Germany)	Crowns
	In-Ceram Zirconia (VITA, Zahnfabrik, Germany)	Posterior crowns
	Lava (3M ESPE, USA) (Partially stabilized zirconia)	Crowns, bridgework
	Procera (Nobel Biocare AB, Sweden) (Partially stabilized zirconia)	Implant abutments
	Procera (Al_2O_3) (Nobel Biocare AB, Sweden)	Implant abutments

pattern invested in a phosphate-bonded investment material. The crown was broken out of the investment and fitted back onto the die. Application of a surface tint and re-firing was used to customize the crown's appearance and ensure the correct shade (Figure 11.10). This type of restoration was made on a conventional all-ceramic crown preparation (see Figure 11.6) but the wearing and chipping of the surface characterization meant that the aesthetics of this restoration could not be assured long term.

Hot-pressed and injection-moulded ceramics

The lost wax technique was also applied to hot-pressed and injection-moulded ceramics. Ivoclar produced a heat-pressed leucite-reinforced glass ceramic to develop the IPS Empress crown (Figure 11.11). Further developments using lithium disilicate glass ceramics gave rise to the IPS Empress 2 and the IPS e.max crowns demonstrating increased strength, thus extending the application of

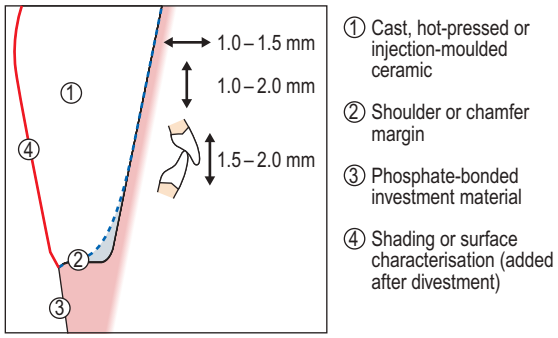


Figure 11.10 Schematic diagram of castable glass, hot-pressed or injection-moulded ceramic crown.



Figure 11.11A Empress crowns placed on upper central incisor teeth. The tooth preparation can be seen inset.



Figure 11.11B Empress crowns placed on the central incisors and upper right lateral incisor. Excellent aesthetic results are possible with such restorations.

such crowns to premolar teeth. In common with cast glass, the surface is characterized with tints although the basic shade has been developed in hot-pressed and injection-moulded glass (see Figure 11.10).

Development of a generic all-ceramic crown preparation

With the further development of alternative materials for all-ceramic crowns, there was increasing demand to develop such restorations for use with posterior teeth. Accordingly, preparations for posterior all-ceramic crowns were developed and generically comprise between 1.0 and 1.5 mm axial reduction, 1.0–2.0 mm incisal edge or occlusal reduction, and between 1.5 and 2.0 mm interocclusal clearance for both anterior and posterior teeth. Such a preparation is outlined diagrammatically in Figure 11.12. Generally, advances in materials have resulted in more minimal tooth reduction and the development of a chamfer margin in preference to a rounded shoulder.

Development of improved coping materials

In order to increase further the strength of all-ceramic crowns, attention focussed on the development of a stronger coping covered with feldspathic porcelain. VITA developed a group of novel coping materials created by building up a coping on a refractory die which shrinks as the coping is fired, thus making it very easy to remove from the die (Figure 11.13). These glass-infused coping materials are:

- Alumina, as the basis for the In-Ceram Alumina crown
- Alumina with added magnesia as the basis for the In-Ceram Spinell crown
- A combination of alumina and zirconia as the basis for the In-Ceram Zirconia crown.

Glass-infused ceramic copings

The basis of many of these is a refractory die of the tooth preparation upon which either an almost pure alumina (Al_2O_3) (In-Ceram Alumina, VITA, Zahnfabrik) or less hard, but of greater translucency, magnesia and alumina

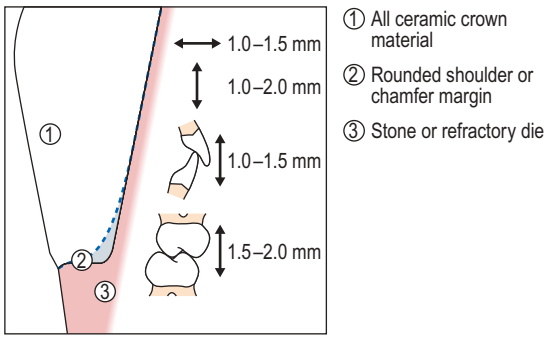


Figure 11.12 The design of a modern all-ceramic crown.

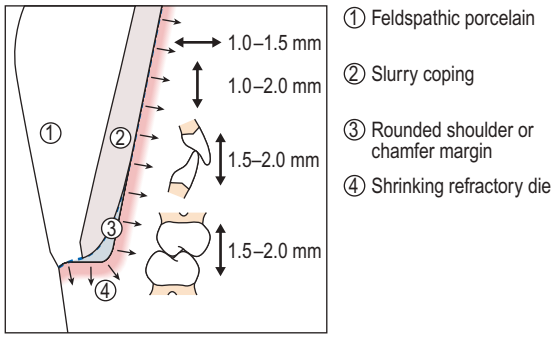


Figure 11.13 Schematic diagram of an In-Ceram all-ceramic crown with a strengthened coping.

($MgAl_2O_4$) (In-Ceram Spinell, VITA, Zahnfabrik) coping may be built up. Following sintering (at around 1200°C for 10 hours) the resultant structure is not structurally strong, so a slurry of lanthanum glass is painted on. Upon re-firing (at around 1100°C for 4 hours) the applied glass flows into the voids of the underlying coping, strengthening it.

The completed coping may then be built upon with suitable feldspathic porcelain. With regard to this the coefficient of thermal expansion of the coping and covering porcelain must be the same if cracking of the porcelain is to be avoided during the firing process. It should also be noted that such copings may also be milled from blanks of these glass-infused ceramic core materials using CAD-CAM technology (see later).

THE USE OF CAD-CAM AND DEVELOPMENT OF GLASS CERAMIC MATERIALS WITH SIGNIFICANTLY INCREASED STRENGTH

CAD-CAM has significantly changed all ceramic restorations with respect to:

- The types of material that can be used for the manufacture of all-ceramic crown copings
- The way dies are used and produced
- The way in which crowns can now be made without a master die model.

The advent of computer-aided manufacture to control milling machines has facilitated the use of extremely strong ceramic materials such as pure alumina or zirconium or yttrium-stabilized zirconium to form copings which fit over the prepared die and support more aesthetic feldspathic porcelain.

Nobel Biocare developed a technique whereby a stone die from an impression taken in the clinic is scanned using a laser to produce a digital three-dimensional map. This information can be sent electronically to a milling machine, often many miles away, for computer-aided manufacture of alumina and, more recently, zirconium copings. The copings are returned by post to the laboratory where they are fitted back onto the stone die prior to the addition of aesthetic feldspathic porcelain. This is very important as alumina, and particularly zirconium, is often a dense white colour which can be difficult to disguise with porcelain (Figure 11.14). In this way, Procera crowns are constructed. These crowns have significantly increased strength and have therefore been prescribed for the restoration of molar teeth. The manufacturer claims compressive strength in the region of 500 MPa. The fitting surface of the coping has a similar surface roughness to etched porcelain which assists the bonding process to the underlying core.



Figure 11.14 Zirconia coping (Lava) and completed anterior crown.

3M ESPE also developed a CAD-CAM method for milling zirconium in its green state followed by a sintering process for which account is taken of the three-dimensional shrinkage of the zirconium (Figure 11.15). This extremely hard and strong coping is then fitted back onto a stone die for the technician to build up the aesthetic feldspathic porcelain superstructure, disguising the relatively unaesthetic zirconium coping. 3M ESPE developed eight basic shades of zirconium for milling which has helped to improve the appearance of what are known as Lava crowns. The manufacturer's claim for the compressive strength of this type of crown is between 700 and 800 MPa.

VITA In-Ceram have used CAD-CAM to develop a set of copings from alumina, zirconium and yttrium-stabilized zirconium with manufacturer-claimed compressive strengths of 500, 600 and 900 MPa, respectively. In common with previously described copings, an aesthetic layer of feldspathic-like porcelain is placed over the coping by the technician.

Long-term studies of all-ceramic crowns manufactured using a CAD-CAM coping covered by a layer of aesthetic porcelain are scarce. However, those that do exist indicate a potential risk of loss of unsupported feldspathic porcelain from the underlying coping, a process called delamination.



Figure 11.15 The Lava milling machine. The white rectangular zirconia blanks can be seen stacked up to the left, the milling machine is to the right.
(Courtesy of 3M ESPE)

Early copings were designed as 'caps' to cover the prepared tooth. They did not provide support for the overlaid porcelain which is weaker and of uneven thickness. By designing the coping to reflect the shape of what would have been underlying dentine, and using the feldspathic veneering porcelain to represent enamel for the final shape of the completed crown, better support should be provided, thereby reducing the risk of delamination. CAD-CAM technology allows accurate design of these 'biologic' copings.

The use of an intraoral camera to capture a series of images resulting in a three-dimensional composite image has also been developed to produce a so-called optical impression. The cameras require a dry preparation, often best achieved using rubber dam isolation, covered with a non-reflective opaque spray. In this way, the camera and dedicated computer software can identify and build up a dimensionally accurate image which is used for computer-aided design (CAD) of the restoration prior to computer-aided manufacture (CAM) from a single block of material.

Experience suggests quite a steep learning curve to develop this method of optical impression taking and chairside restoration design for all-ceramic restorations. The Sirona Company initially produced the Cerec machine for this purpose but initial results for marginal fit of the restorations were considered to be suboptimal. Development of the process and the advent of Cerec III has resolved this issue and excellent restorations can now be made.

A recent development by 3M ESPE is the Lava chairside oral scanner (COS) whereby a full arch can be scanned by moving a video camera around the arch (Figure 11.16).



Figure 11.16 The chairside oral scanner (COS). The intraoral scanner is on the right side mount.
(Courtesy of 3M ESPE)

In addition, a three-dimensional image of the opposing dentition ensures restorations can be designed, on virtual dies, without occlusal interferences. Furthermore, the use of a technique called stereolithography (SLA) can be used to produce consistently accurate and durable, resin, occluding models and dies for laboratory construction of conventional extra- and intracoronal restorations such as gold shell crowns and porcelain bonded to metal crowns, in addition to all-ceramic crowns. The SLA model can also be used to ensure the fit of an anatomically accurate milled coping for subsequent addition of porcelain.

Blocks of material from which entire crowns can be milled are manufactured by VITA and include VITA Mark II ceramic blocks where the final shade of the restoration is developed by light scattering from any surrounding tooth structure, thus creating a chameleon effect. Coloured VITA TriLuxe and TriLuxe Forte ceramic blocks reproduce the more intense shading at the gingival margin with the lighter colour of the cusp tip or incisal edge.

Blocks of leucite-reinforced ceramic can also be used (Empress CAD, IPS e.max CAD and Paradigm-C) which come as multichromatic blocks or in five monoshade blocks. In each case, surface tints are added and fired in the laboratory. For the Empress CAD crowns, compressive strengths of 160 MPa are claimed. The IPS e.max CAD crowns are milled in the 'soft' state prior to a short 30-minute crystallization process to produce restorations with a compressive strength of 320 MPa.

3M ESPE have also developed a block based on their Z100 composite (Paradigm MZ100) which comes in five basic shades, claims enamel-like wear characteristics, can be characterized at the chairside with light-cured stains and can be repaired intraorally if required.

CEMENTATION OF PORCELAIN RESTORATIONS

Due to both the aesthetic requirements and brittle nature of some ceramic restorations their cementation poses a challenge. In the case of veneers they must first be tried in, not only to assess their fit but also their appearance. Their aesthetics, by virtue of thinness and translucency, can be markedly affected by the shade of the underlying cement. Resin composite luting agents are used to achieve final cementation and such products are specially formulated for this purpose.

Each manufacturer produces a range of trial pastes that may be applied to the fit surface of the crown or veneer at try in. These are of shades that match the final luting cements. Once this has been carried out the veneer is readily removed and both the tooth preparation and veneers must be debrided of residual paste prior to final cementation of the restoration. This is readily accomplished by a prophylaxis of the tooth and steam cleaning of the veneer. Cementation of ceramic veneers will be discussed in further in Chapter 12.

In relation to final cementation of all porcelain restorations it is important to note that the materials used for this should exhibit shear thinning (a reduction in viscosity with applied pressure) to minimize fracture of the restoration upon seating. The cementing medium should also provide a firm foundation for the restoration to maximize resistance to fracture. Acid-based conventional cements (e.g. zinc oxyphosphate, zinc polycarboxylate and conventional glass polyalkenoate) should be avoided as they may interact with porcelain, exacerbating any flaws and enhancing crack propagation in function. As a consequence, resin composite luting agents are recommended. Such materials are not acid based and exhibit superior flexural strength. Those that are dual cured are considered best, since lone chemical curing lacks operator control and satisfactory light curing alone may not be accomplished if the restoration is either deep in dimension or dark in colour.

The exception to this choice of cement type is zirconia oxide-based crowns that may, if desired, be cemented with conventional cements. Their high fracture resistance diminishes the need for support and their underlying tooth preparation requires no adhesive interface for retention.

The fit surface of all feldspathic porcelain restorations, to be cemented with resin composite luting agents, must be etched prior to cementation with hydrofluoric acid (5–9.5%). This is best accomplished in the dental laboratory. As an alternative, air abrasion is to be avoided as this will induce and exacerbate any flaws in the porcelain. The fit surface of the restoration should then be silanated just prior to the application of the resin composite luting agent. It is important to realize that silane is a chemical coupling agent that will bond the porcelain to the luting agent. This chemical exhibits good wetting but degrades in the presence of both moisture and air. Its application should therefore be as near as possible to the cementation of the restoration. Bonding of the resin luting agent to tooth substance is achieved by acid etching the preparation, rinsing, drying and applying a bonding agent prior to seating the resin composite loaded restoration.

Glass infiltrated alumina-based ceramics are resistant to hydrofluoric acid etching and, in the case of restorations fabricated from this material, one manufacturer (3M ESPE) advocates a tribochemical silica coating process instead. This roughens the fit surface to facilitate bonding of the resin composite luting agent. This is accomplished by laboratory cleaning of the restoration by application of 100 μm high purity aluminium oxide under a pressure of 250 kPa for 14 seconds. A similar process pertains for sintered aluminium oxide crown copings.

SUMMARY

The long history of all-ceramic crowns has utilized advances in ceramic materials and production methods to create functional and aesthetic restorations which require increasingly less tooth reduction to achieve an adequate result. On-going clinical studies will inform greater understanding about the modes of failure of these restorations and help to further develop materials and techniques, thus extending the range of clinical circumstances in which this type of restoration may be used. Many of the clinical studies reporting the results of all-ceramic crowns indicate that they are successful for anterior teeth. The situation on posterior teeth, either single crowns or bridges, is more complex. Small fractures or chips are commonly found and whilst these may not result in failure of the restoration, they are unwanted complications. As such, these restorations should be carefully considered for patients using high occlusal loading, particularly bruxists. The balance between the need to achieve a good aesthetic result and a long-term successful restoration means that these restorations need careful planning and prescription.

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